Data Structures and Algorithms COMS2004A

Introduction and Revision of C++

Contact Information

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^(This is all in the CBO!)

CBO

Introduction

What are “Data Structures” and “Algorithms”?

Data Structures? 🤔

Algorithms? 🤔

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- Examples of use:
  - Relational databases, networking and routing, operating systems, version control, streaming systems… the list goes on and on!

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- Examples of use:
  - Sorting and searching, GPS route finding, data compression, encryption, machine learning… there is also a lot!

Data Structures & Algorithms

- Organised ways to store data
- Provide the step-by-step logic to operate on the data

Why Data Structures and Algorithms? Why should I care

We have realised now that these concepts are ubiquitous/foundational in computers, computer engineering and computer science.

But, how did this come to be?

Why Data Structures and Algorithms? Why should I care

One problem can be solved in many ways. But:

- Not all ways are equal. The difference between a good and a bad choice can be nanoseconds or milliseconds vs. days or weeks.
- Hardware gets faster, but data grows faster.

  Efficiency comes from **how** we compute, not just what we compute on.

- They also give us a shared language and system for reasoning about **cost/computational work** before we write code.

Why Data Structures and Algorithms? Why should I care

In short:

- They are fundamental building blocks or efficient, scalable software.

C++ Revision

(VIP, but self-study. This is prerequisite)

C++ Refresher Topics (please self-study - check Moodle resources)

Your C++ basics:

- Fundamental types
- Pointers, arrays and structures
- Named constants, scope and namespaces
- Expressions
- Casting
- Control Flow
- Functions
- Classes and OOP

C++ Refresher Example Demo

- Someone asks you to check if a string is an anagram of another
- Anagram: a word formed by rearranging letters of another
  - For simplicity's sake in this example, let's say we must be case-sensitive
  - E.g., "debit card" and "bad credit" are anagrams